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Power efficient crest factor reduction

Abstract

Crest factor reduction circuitry includes: a peak neighborhood analyzer; peak detection circuitry and a controller. The peak neighborhood analyzer is configured to: receive an input signal; analyze the input signal to determine whether a peak larger than a target threshold is expected within an interval; and provide a first control signal responsive to determining that a peak larger than the target threshold is expected within the interval. The controller is configured to: receive the first control signal; and gate a clock or data to the peak detection circuitry responsive to the first control signal.

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to India Provisional Application No. 202341031976, titled “LOW POWER ARCHITECTURE FOR CREST FACTOR REDUCTION”, filed on May 5, 2023, which is hereby incorporated by reference in its entirety.

BACKGROUND

(2) An example base station has a transmitter and a power amplifier. If the transmit signal from the transmitter has a high peak-to-average power ratio (PAR), the output root-mean-square (RMS) power and the efficiency of the power amplifier is reduced. Such power amplifiers are non-linear and may be affected by out-of-band emissions and transmission spectral mask violations. To improve linearity of power amplifiers, the transmitter may condition a transmit signal using, for example, Crest Factor Reduction (CFR) and Digital Pre-Distortion (DPD), where CFR reduces the Peak-to-Average power Ratio (PAR) of the transmit signal prior to DPD. Although CFR can improve a power amplifier's dynamic range and increase power amplifier efficiency and RMS power, the power consumption of CFR operations can be significant.

SUMMARY

(3) In an example, a circuit includes crest factor reduction circuitry. The crest factor reduction (CFR) circuitry includes: a peak neighborhood analyzer having a first terminal and a second terminal; peak detection circuitry having a first terminal, a second terminal, and a third terminal; and a controller having a first terminal and a second terminal. The first terminal of the controller is coupled to the second terminal of the peak neighborhood analyzer. The second terminal of the controller is coupled to the second terminal of the peak detection circuitry. The peak neighborhood analyzer is configured to: receive an input signal at the first terminal of the peak neighborhood analyzer; analyze the input signal to determine whether a peak larger than a target threshold is expected within an interval; and provide a first control signal at the second terminal of the peak

neighborhood analyzer responsive to determining that a peak larger than the target threshold is expected within the interval. The controller is configured to: receive the first control signal at the first terminal of the controller; and gate a clock or data to the peak detection circuitry responsive to the first control signal.

(4) In another example, a transmitter includes CFR circuitry. The CFR circuitry includes: peak detection circuitry having a first terminal, a second terminal, and a third terminal; and peak cancellation having a first terminal, a second terminal, and a third terminal. The first terminal of the peak cancellation circuitry is coupled to the third terminal of the peak detection circuitry. The peak cancellation circuitry includes: envelope calculation circuitry; excess peak calculation circuitry; a delay line; and a multiplier. The envelope calculation circuitry has a first terminal and a second terminal. The first terminal of the envelope calculation circuitry is coupled to the first terminal of the peak cancellation circuitry. The excess peak calculation circuitry has a first terminal and a second terminal. The first terminal of the excess peak calculation circuitry is coupled to the second terminal of the envelope calculation circuitry. The delay line has a first terminal and a second terminal. The first terminal of the delay line is coupled to the first terminal of the peak cancellation circuitry. The multiplier has a first terminal, a second terminal, and a third terminal. The first terminal of the multiplier is coupled to the second terminal of the delay line. The second terminal of the multiplier is coupled to the second terminal of the excess peak calculation circuitry.

(5) In yet another example, a method includes: receiving a transmit signal; selectively enabling peak detection based on a peak absence prediction for the transmit signal; when enabled, performing peak detection by comparing a peak of the transmit signal to a peak limit; calculating an excess peak portion of the peak relative to the peak limit; applying a correction to the peak based on the excess peak portion; and providing an updated transmit signal with the corrected peak.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a block diagram showing an example circuit.

(2) FIG. **2** is a diagram showing another example circuit.

(3) FIG. **3** is a diagram showing example Crest Factor Reduction (CFR) circuitry.

(4) FIG. **4** is a diagram showing another example CFR circuitry.

(5) FIG. **5** is a graph showing a complex envelope complementary cumulative distribution function (CCDF) of an orthogonal frequency division multiplexing (OFDM) signal.

(6) FIG. **6**A is a graph showing an example input signal envelope, a peak absence prediction threshold, and a peak limit.

(7) FIG. **6**B is a diagram showing an example complementary cumulative distribution function (CCDF) of a peak-to-average ratio (PAR) for a CFR input and a CFR output.

(8) FIG. **7** is a block diagram of example cancellation phasor control circuitry.

(9) FIGS. **8**, **9**, and **10** are flowcharts of example CFR methods.

DETAILED DESCRIPTION

(10) The same reference numbers or other reference designators are used in the drawings to designate the same or similar features. Such features may be the same or similar either by function and/or structure.

(11) Described herein are power efficient Crest Factor Reduction (CFR) topologies and control options. In some examples, CFR circuitry may include one or more CFR stages. Each CFR stage includes a peak neighborhood analyzer, interpolated envelope computation circuitry, peak detection circuitry, and peak cancellation circuitry. The peak neighborhood analyzer operates to: determine whether there are any significant peaks within an upcoming interval, window, or related number of input signal samples (sometimes referred to as transmit signal samples herein); and provide a

control signal responsive to determining that there is no significant peak within the upcoming interval, window, or related number of input signal samples. In some examples, the input signal (sometimes referred to as a transmit signal herein) may be a 4G or 5G signal based on orthogonal frequency division multiplexing (OFDM) modulation.

(12) In the described examples, different techniques to determine the presence or absence of significant peaks may be used by the peak neighborhood analyzer. In some examples, a significant peak is identified by: calculating envelope values for input signal samples for a baseband interface rate; and comparing the envelope values to a threshold (sometimes referred to herein as a peak absence prediction threshold) to obtain comparison results. If the comparison results indicate a number of consecutive envelope values are below the threshold, no significant peak within the upcoming interval, window, or related number of input signal samples is assumed and the control signal is provided.

(13) In response to the control signal being provided by the peak neighborhood analyzer, the interpolated envelope computation circuitry and/or the peak detection circuitry are disabled. For example, data and/or clocks for the interpolated envelope computation circuitry and/or the peak detection circuitry may be gated temporarily based on the interval, window, or related number of input signal samples. If the peak neighborhood analyzer determines that there is a significant peak within the interval, window, or the related number of input signal samples, the interpolated envelope computation circuitry and the peak detection circuitry are not disabled and operate to determine whether there are any peaks greater than a peak limit. If a detected peak is greater than the peak limit, the peak cancellation circuitry applies a correction so that the detected peak is reduced to below the peak limit. In some examples, the peak cancellation circuitry operates to: determine an excess peak portion of a detected peak relative to the peak limit; scale a complex baseband interpolated sample y.sub.i corresponding to the peak responsive to the excess peak portion to generate a cancellation phasor; generate a peak cancellation pulse responsive to the cancellation phasor and a peak cancellation waveform; and add the peak cancellation pulse to a latency (delay) matched version of the input signal to apply peak correction or limiting.

(14) In some examples, multiple stages of CFR circuitry (e.g., up to 3 stages) may be used to perform peak correction or peak limiting operations. Each stage may include a respective peak neighborhood analyzer, respective interpolated envelope computation circuitry, respective peak detection circuitry, and respective peak cancellation circuitry. With the described peak neighborhood analyzer and/or the described peak cancellation circuitry, power consumption of CFR operations is reduced relative to other CFR techniques. In some examples, application-specific hardware may be used to perform each of the CFR operations described herein. In other examples, at least some of the CFR operations may be performed using a processor and a memory that stores related instructions.

(15) In FIGS. **3**, **4**, and **7**, example CFR circuitry is described. Such CFR circuitry may be categorized as digital signal processing/conditioning circuitry. In some examples, the example CFR circuitry of FIGS. **3**, **4**, and **7** is implemented using digital hardware. Example digital hardware that may be used for CFR circuitry includes combinational logic (e.g., adders/subtractors, multipliers, counters, comparators and logic gates including AND/OR/NOT gates), sequential logic (e.g., flip-flops, latches, delay chains), and/or memory elements (e.g., random-access memory (RAM) elements and/or read-only memory (ROM) elements). The digital hardware used for CFR circuitry may be part of one or more integrated circuits (ICs). As another option, digital hardware for CFR circuitry may be synthesized and implemented using Flexible Programmable Gate Arrays (FPGAs). As another option, processor/firmware/software elements may be used to perform some or all CFR operations. Without limitation, dedicated digital hardware may be preferred over processor/firmware/software elements depending on the target speed of individual and/or collective CFR operations.

(16) FIG. **1** is a block diagram showing an example circuit **100**. In some examples, the circuit **100**

is part of the circuitry of a base station circuit or data transmission device. As shown, the circuit **100** includes a transmitter **102** having a first terminal **104** and a second terminal **106**. The transmitter **102** includes CFR circuitry **108**. The CFR circuitry **108** includes a peak neighborhood analyzer **110** and cancellation phasor control circuitry **114**. In some examples, the CFR circuitry **108** may also include other components such as interpolated envelope computation circuitry, peak detection circuitry, and/or peak cancellation circuitry (see e.g., the components of FIGS. **3**, **4**, and **7**). In some examples, the CFR circuitry **108** includes multiple CFR stages (e.g., up to 3 stages), where each CFR stage includes a respective peak neighborhood analyzer, respective interpolated envelope computation circuitry, respective peak detection circuitry, and respective peak cancellation circuitry.

(17) The peak neighborhood analyzer **110** operates to: determine whether there are any significant peaks within an upcoming interval of an input signal sample; and provide a control signal responsive to determining that there is no significant peak within the interval. If there is a significant peak within the interval, the CFR circuitry **108** operates to: detect whether the significant peak is greater than a peak limit. In some examples, the peak neighborhood analyzer **110** includes peak absence prediction circuitry **112** to determine whether there are any significant peaks within the interval. In some examples, the peak absence prediction circuitry **112** operates to: obtain input signal samples related to the interval; calculate envelope values of the input signal samples, compare the input signal samples to a significant peak pattern or threshold to obtain comparison results; and provide the control signal based on the comparison results indicating there is no significant peak within the interval.

(18) If a detected peak is greater than the peak limit, the cancellation phasor control circuitry **114** operates to: determine an excess peak portion of a detected peak relative to the peak limit; scale a sample related to the peak responsive to the excess peak portion to generate a cancellation phasor. In some examples, the cancellation phasor is used to scale a peak cancellation waveform to generate a peak cancellation pulse. The peak cancellation pulse is added to a latency (delay) matched version of the input signal to apply peak correction or limiting. In some examples, the CFR circuitry **108** repeats the same or similar CFR operations for each of multiple CFR stages.

(19) In some examples, the cancellation phasor control circuitry **114** includes excess peak scaling circuitry **116** that operates to: determine an excess peak portion of a detected peak; receive a complex baseband interpolated sample y.sub.i corresponding to the detected peak; and scale the complex baseband interpolated sample responsive to the excess peak portion to generate a cancellation phasor. In some examples, the excess peak scaling circuitry **116** uses a look-up table (LUT) indexed using a detected peak sample to expedite determining the excess peak portion of the detected peak relative to the peak limit. Other CFR options and details are described hereafter.

(20) With the CFR circuitry **108**, the transmitter **102** receives an input signal at the first terminal **104**, where the input signal may include peaks greater than the peak limit. As part of conditioning the input signal for subsequent transmission operations, the transmitter **102** uses the CFR circuitry **108** to reduce the amplitude of peaks to below the peak limit. In different examples, the peak limit may vary to achieve a target Peak-to-Average power Ratio (PAR) of a baseband signal prior to other operations of the transmitter **102**.

(21) FIG. **2** is a diagram showing another example circuit **200**. As shown, the circuit **200** includes a processor **202**, transmitter circuitry **208**, power amplifier circuitry **270**, and an antenna **290**. The processor **202** has a first terminal **204** and a second terminal **206**. The transmitter circuitry **208** has a first terminal **210**, a second terminal **212**, a third terminal **214**, and a fourth terminal **216**. The power amplifier circuitry **270** has a first terminal **272**, a second terminal **274**, and a third terminal **276**.

(22) The transmitter circuitry **208** includes CFR circuitry **108**A, interpolation circuitry **220**, Digital Pre-Distortion (DPD) corrector circuitry **224**, DPD estimator circuitry **230**, digital circuitry **238**, a digital-to-analog converter (DAC) **244**, a digital step attenuator (DSA) **250**, a feedback analog-to-

digital converter (ADC) **256**, and feedback digital circuitry **262**. The CFR circuitry **108**A is an example of the CFR circuitry **108** in FIG. **1**. In the example of FIG. **2**, some or all of the transmitter circuitry **208** is included in the transmitter **102** of FIG. **1**. As shown, the CFR circuitry **108**A has a first terminal **218** and a second terminal **219**. The interpolation circuitry **220** has a first terminal **221** and a second terminal **222**. The DPD corrector circuitry **224** has a first terminal **226**, a second terminal **227**, and a third terminal **228**. The DPD estimator circuitry **230** has a first terminal **232**, a second terminal **234**, a third terminal **235**, and a fourth terminal **236**. The digital circuitry **238** has a first terminal **240** and a second terminal **242**. The DAC **244** has a first terminal **246** and a second terminal **248**. The DSA **250** has a first terminal **252** and a second terminal **254**.

(23) In the example of FIG. **2**, the power amplifier circuitry **270** includes a power amplifier **278** and switch/diplexer circuitry **284**. The power amplifier **278** has a first terminal **280** and a second terminal **282**. The switch/diplexer circuitry **284** has a first terminal **286** and a second terminal **288**.

(24) In the example of FIG. **2**, the first terminal **204** of the processor **202** is coupled to the first terminal **210** of the transmitter circuitry **208**. The second terminal **206** of the processor **202** is coupled to the second terminal **212** of the transmitter circuitry **208**. The third terminal **214** of the transmitter circuitry **208** is coupled to the first terminal **272** of the power amplifier circuitry **270**. The fourth terminal **216** of the transmitter circuitry **208** is coupled to the third terminal **276** of the power amplifier circuitry **270**. The second terminal **274** of the power amplifier circuitry **270** is coupled to the antenna **290**.

(25) The first terminal **210** of the transmitter circuitry **208** is coupled to the first terminal **218** of the CFR circuitry **108**A. The second terminal **219** of the CFR circuitry **108**A is coupled to the first terminal of the interpolation circuitry **220**. The second terminal **222** of the interpolation circuitry **220** is coupled to the first terminal **226** of the DPD corrector circuitry **224** and to the second terminal **234** of the DPD estimation circuitry **230**. The second terminal of the DPD corrector circuitry **224** is coupled to the fourth terminal **236** of the DPD estimation circuitry **230**. The third terminal **228** of the DPD corrector circuitry **224** is coupled to the first terminal **240** of the digital circuitry **238** and to the first terminal **232** of the DPD estimation circuitry **230**. The second terminal **242** of the digital circuitry **238** is coupled to the first terminal **246** of the DAC **244**. The second terminal **248** of the DAC is coupled to the first terminal **252** of the DSA **250**. The second terminal **254** of the DSA **250** is coupled to the third terminal **214** of the transmitter circuitry **208**. The fourth terminal **216** of the transmitter circuitry **208** is coupled to the first terminal **258** of the feedback ADC **256**. The second terminal **260** of the feedback ADC **256** is coupled to the first terminal **264** of the feedback digital circuitry **262**. The second terminal **266** of the feedback digital circuitry **262** is coupled to the second terminal **212** of the transmitter circuitry **208** and to the third terminal **235** of the DPD estimator circuitry **230**.

(26) The first terminal **272** of the power amplifier circuitry **270** is coupled to the first terminal **280** of the power amplifier **278**. The second terminal **282** of the power amplifier **278** is coupled to the third terminal **276** of the power amplifier circuitry **270** and to the first terminal **286** of the switch/diplexer circuitry **284**. The second terminal **288** of the switch/diplexer circuitry **284** is coupled to the second terminal **274** of the power amplifier circuitry **270**.

(27) The processor **202** operates to: prepare a transmit baseband signal; and provide the transmit baseband signal to the first terminal **204** responsive to a particular transmission protocol. In some examples, the transmit baseband signal is a 4G or 5G signal based on OFDM modulation. In other examples, the transmit baseband signal may be a complex baseband signal corresponding to other transmission standards, such as wireless local-area network (WLAN). In some examples, the transmit baseband signal provided by the processor **202** is based in part on feedback **268** received at the second terminal **206** of the processor **202**. In other examples, the feedback **268** to the processor **202** is not used and may be omitted.

(28) The transmitter circuitry **208** operates to: receive the transmit baseband signal at the first terminal **210**; condition the transmit baseband signal using the CFR circuitry **108**A; the

interpolation circuitry **220**, the DPD estimator circuitry **230**, the DPD corrector circuitry **224**, and the digital circuitry **238**; convert the conditioned transmit signal to an analog transmit signal using the DAC **244**; apply a gain or attenuation to the analog transmit signal using the DSA **250**; and provide the buffered analog transmit signal to the third terminal **214**. The transmitter circuitry **208** also operates to receive the power amplifier output at the fourth terminal **216**; digitize the power amplifier output using the feedback ADC **256**; adjust the digitized power amplifier output using the feedback digital circuitry **262**; and provide the adjusted digitized power amplifier output to the second terminal **212**. In some examples, the CFR circuitry **108**A performs the operations described for the CFR circuitry **108**B in FIG. **3**. In some examples, the interpolation circuitry **220** includes a chain of filters that operate to increase the sampling rate of the received signal from an input sampling rate to a higher sampling rate for use by the DPD estimator circuitry **230** and the DPD corrector circuitry **224**. With the transmitter circuitry **208** of FIG. **2**, power consumption of CFR operations is reduced responsive to the operations of the peak neighborhood analyzer **110** and the cancellation phasor control circuitry **114** described in FIG. **1**. Additional CFR options and details are described herein.

(29) The power amplifier circuitry **270** operates to: receive the buffered analog transmit signal at the first terminal **272**; amplify the buffered analog transmit signal using the power amplifier **278**; selectively provide the amplified transmit signal to the second terminal **274**; and provide the amplified transmit signal to the third terminal **276**.

(30) FIG. **3** is a diagram showing example CFR circuitry **108**B. The CFR circuitry **108**B is an example of the CFR circuitry **108** in FIG. **1** and the CFR circuitry **108**A in FIG. **2**. In the example of FIG. **3**, the CFR circuitry **108**B has the first terminal **218** and the second terminal **219** described in FIG. **2**. In some examples, the CFR circuitry **108**B includes interpolation circuitry **302**, interpolated envelope calculation circuitry **310**, peak detection circuitry **318**, a peak neighborhood analyzer **110**A, peak cancellation circuitry **380**, and additional CFR stages **372**.

(31) The peak neighborhood analyzer **110**A is an example of the peak neighborhood analyzer **110** of FIG. **1**. In the example of FIG. **3**, the peak neighborhood analyzer **110**A includes envelope calculation circuitry **330** and peak absence prediction circuitry **112**A. The peak absence prediction circuitry **112**A is an example of the peak absence prediction circuitry **112**A in FIG. **1**. The cancellation phasor control circuitry **114**A is an example of the cancellation phasor control circuitry **114** of FIG. **1**.

(32) As shown, the interpolation circuitry **302** has a first terminal **304**, a second terminal **306**, and a third terminal **308**. The interpolated envelope calculation circuitry **310** has a first terminal **312**, a second terminal **314**, and a third terminal **316**. The peak detection circuitry **318** has a first terminal **320**, a second terminal **322**, and a third terminal **324**. The peak neighborhood analyzer **110**A has a first terminal **326** and a second terminal **328**. The envelope calculation circuitry **330** has a first terminal **332** and a second terminal **334**. The peak absence prediction circuitry **112**A has a first terminal **338** and a second terminal **340**.

(33) The peak cancellation circuitry **380** has a first terminal **382**, a second terminal **384**, and a third terminal **386**. In the example of FIG. **3**, the peak cancellation circuitry **380** includes the cancellation phasor control circuitry **114**A, cancellation pulse generation circuitry **345**, delay circuitry **356**, and combine circuitry **364**. The cancellation phasor control circuitry **114**A includes envelope calculation circuitry **346** and excess peak scaling circuitry **116**A. The excess peak scaling circuitry **116**A is an example of the excess peak scaling circuitry **116** in FIG. **1**.

(34) The cancellation phasor control circuitry **114**A has a first terminal **342** and a second terminal **344**. The envelope calculation circuitry **346** has a first terminal **348** and a second terminal **350**. The excess peak scaling circuitry **116**A has a first terminal **352** and a second terminal **354**. The cancellation pulse generation circuitry **345** has a first terminal **347** and a second terminal **349**. The delay circuitry **356** has a first terminal **358**, a second terminal **360**, and a third terminal **362**. The combine circuitry **364** has a first terminal **366**, a second terminal **368**, and a third terminal **370**. The

additional CFR stages **372** have a first terminal **374** and a second terminal **376**.

(35) In the example of FIG. **3**, the first terminal **218** of the CFR circuitry **108**B is coupled to the first terminal **304** of the interpolation circuitry **302**, the first terminal **326** of the peak neighborhood analyzer **110**A, and the second terminal **384** of the peak cancellation circuitry **380**. The second terminal **306** of the interpolation circuitry **302** is coupled to the second terminal **328** of the peak neighborhood analyzer **110**A. The third terminal **308** of the interpolation circuitry **302** is coupled to the first terminal **312** of the interpolated envelope calculation circuitry **310**. The second terminal **314** of the interpolated envelope calculation circuitry **310** is coupled to the second terminal **328** of the peak neighborhood analyzer **110**A. The third terminal **316** of the interpolated envelope calculation circuitry **310** is coupled to the first terminal **320** of the peak detection circuitry **318**. The second terminal **322** of the peak detection circuitry **318** is coupled to the second terminal **328** of the peak neighborhood analyzer **110**A. The third terminal **324** of the peak detection circuitry **318** is coupled to the first terminal **382** of the peak cancellation circuitry **380**.

(36) In the example of FIG. **3**, the first terminal **326** of the peak neighborhood analyzer **110**A is coupled to the first terminal **332** of the envelope calculation circuitry **330**. The second terminal **334** of the envelope calculation circuitry **330** is coupled to the first terminal **338** of the peak absence prediction circuitry **112**A. The second terminal **340** of the peak absence prediction circuitry **112**A is coupled to the second terminal **328** of the peak neighborhood analyzer **110**A.

(37) The first terminal **382** of the peak cancellation circuitry **380** is coupled to the first terminal **342** of the cancellation phasor control circuitry **114**A. The first terminal **348** of the envelope calculation circuitry **346** is coupled to the first terminal **342** of the cancellation phasor control circuitry **114**A. The second terminal **350** of the envelope calculation circuitry **346** is coupled to the first terminal **352** of the excess peak scaling circuitry **116**A. The second terminal **354** of the excess peak scaling circuitry **116**A is coupled to the second terminal of the cancellation phasor control circuitry **114**A. The second terminal **344** of the cancellation phasor control circuitry **114**A is coupled to the first terminal **347** of the cancellation pulse generation circuitry **345**. The second terminal **349** of the cancellation pulse generation circuitry **345** is coupled to the first terminal **366** of the combine circuitry **364**. The second terminal **368** of the combine circuitry **364** is coupled to the third terminal **362** of the delay circuitry **356**. The first terminal **358** of the delay circuitry **356** is coupled to the second terminal **384** of the peak cancellation circuitry **380**. In some examples, the second terminal **360** of the delay circuitry **356** receives a delay control signal (DELAY_CTRL).

(38) In some examples, DELAY_CTRL is provided by delay estimation circuitry included with the peak cancellation circuitry and/or the CFR circuitry **108**A. In some examples, DELAY_CTRL is a predetermined signal that accounts for delay of the interpolation circuitry **302**, the interpolated envelope calculation circuitry **310**, the peak detection circuitry **318**, the cancellation phasor control circuitry **114**A, and the cancellation pulse generation circuitry **345**. In other examples, DELAY_CTRL is based on actively monitoring delay of the interpolation circuitry **302**, the interpolated envelope calculation circuitry **310**, the peak detection circuitry **318**, the cancellation phasor control circuitry **114**A, and the cancellation pulse generation circuitry **345**. With DELAY_CTRL, the delay circuitry **356** operates to provide a delayed version of x(m) (the delayed version given as x(m)_DLY herein), where x(m)_DLY is a latency (delay) matched version of x(m) to align the peak of x(m) with z(m) from the cancellation pulse generation circuitry **345**.

(39) As shown, the second terminal **368** of the combine circuitry **364** is coupled to the third terminal **362** of the delay circuitry **356** and receives x(m)_DLY. The third terminal **370** of the combine circuitry **364** is coupled to the first terminal **374** of the additional CFR stages **372**. The second terminal **376** of the additional CFR stages **372** is coupled to the second terminal **219** of the CFR circuitry **108**B.

(40) In some examples, the peak neighborhood analyzer **110**A operates to: receive an input signal x(m) having a baseband interface sampling frequency (f.sub.int) at the first terminal **326**; calculate an envelope value of x(m) using the envelope calculation circuitry **330**; and provide the calculated

envelope value of x(m) to the second terminal **334**. In some examples, the calculated envelope value of x(m) is given as |x(m)|.sup.2. The peak absence prediction circuitry **112**A is configured to: receive the calculated envelope value at the first terminal **338**; compare the calculated envelope value to a threshold to obtain comparison results; and provide a control signal l(m) at the second terminal **328** responsive to the comparison results.

(41) The interpolation circuitry **302** operates to: receive x(m) at the first terminal **304**; receive l(m) at the second terminal **306**; and selectively provide an interpolated signal y(n) at the third terminal **308** responsive to x(m) and l(m), where y(n) has an oversampling frequency f.sub.OS. In some examples, f.sub.OS is 1× to 4× greater than f.sub.int. In some examples, l(m) is a control signal that selectively enables the interpolation circuitry **302** responsive to the operations of the peak neighborhood analyzer **110**A. l(m) may be used directly with circuitry of the interpolation circuitry **302** if enable/disable is available. As another option, l(m) may be provided to data/clock gating control circuitry (e.g., the data/clock gating control circuitry **432** in FIG. **4**) to stop data and/or clocks from being provided to the interpolation circuitry **302**.

(42) The interpolated envelope calculation circuitry **310** operates: receive y(n) at the first terminal **312**; receive l(m) at the second terminal **314**; selectively calculate an envelope value of y(n) (sometimes referred to herein as an interpolated envelope value) responsive to y(n) and l(m); and provide the calculated envelope value of y(n) at the third terminal **316**. In some examples, the calculated envelope value of y(n) is given as |y(n)|.sup.2. In some examples, l(m) is a control signal that selectively enables the interpolated envelope calculation circuitry **310** responsive to the operations of the peak neighborhood analyzer **110**A. l(m) may be used directly with circuitry of the interpolated envelope calculation circuitry **310** if enable/disable is available. As another option, l(m) may be provided to data/clock gating control circuitry (e.g., the data/clock gating control circuitry **432** in FIG. **4**) to stop data and/or clocks from being provided to the interpolated envelope calculation circuitry **310**.

(43) The peak detection circuitry **318** operates to: receive the envelope value of y(n) at the first terminal **320**; receive l(m) at the second terminal **322**; selectively compare the envelope value of y(n) to a threshold to obtain comparison results responsive to the envelope value of y(n) and l(m); and provide the comparison results at the third terminal **324**. In some examples, the comparison results include a detected peak y.sub.i. In some examples, l(m) is a control signal that selectively enables the peak detection circuitry **318** responsive to the operations of the peak neighborhood analyzer **110**A. l(m) may be used directly with circuitry of the interpolated envelope calculation circuitry **310** if enable/disable is available. As another option, l(m) may be provided to data/clock gating control circuitry (e.g., the data/clock gating control circuitry **432** in FIG. **4**) to stop data and/or clocks from being provided to the interpolated envelope calculation circuitry **310**.

(44) The peak cancellation circuitry **380** operates to: receive y.sub.i at the first terminal **382**; calculate an envelope value of y.sub.i using the envelope calculation circuitry **346**; determine an excess peak portion based on the calculated envelope value of y.sub.i and a peak limit using the excess peak scaling circuitry **116**A; scale a complex baseband interpolated sample y.sub.i corresponding to the detected peak responsive to the excess peak portion using the excess peak scaling circuitry **116**A to generate a cancellation phasor; scale a peak cancellation waveform responsive to the cancellation phasor to generate a peak cancellation pulse (labeled z(m) herein) using the cancellation pulse generation circuitry **345**; and add the peak cancellation pulse to x(m)_DLY using the combine circuitry **364** to produce a corrected input signal.

(45) Specifically, the cancellation phasor control circuitry **114**A operates to: receive y.sub.i at the first terminal **342**; calculate an envelope value of y.sub.i using the envelope calculation circuitry **346**; determine an excess peak portion based on the calculated envelope value of y.sub.i and a peak limit using the excess peak scaling circuitry **116**A; scale y.sub.i responsive to the excess peak portion to generate a cancellation phasor using the excess peak scaling circuitry **116**A; and provide the cancellation phasor at the second terminal **344**. In some examples, the cancellation phasor is

given as α.sub.ie.sup.jΦ.sup.i, where e.sup.jΦ.sup.i is a phasor, and α.sub.i is a scaling applied to the phasor.
(46) The cancellation pulse generation circuitry **345** operates to: receive the cancellation phasor at the first terminal **347**; scale a peak cancellation waveform responsive to the cancellation phasor to generate a peak cancellation pulse z(m); and provide z(m) at the second terminal **349**. In some examples, the peak cancellation waveform is stored in a memory of the cancellation pulse generation circuitry **345** for use in generating z(m) responsive to cancellation phasor. In some examples, the peak cancellation waveform is scaled based on peak attributes (e.g., position, amplitude, phase).
(47) The delay circuitry **356** operates to: receive x(m) at the first terminal **358**; receive DELAY_CTRL at the second terminal **360**; and provided x(m)_DLY at the third terminal **362** responsive to x(m) and DELAY_CTRL. In some examples, DELAY_CTRL is selected or adjusted to account for delays due to the operations of the interpolation circuitry **302**, the peak neighborhood analyzer **110**A, the interpolated envelope calculation circuitry **310**, the peak detection circuitry **318**, and/or the peak cancellation circuitry **380**.
(48) In some examples, the combine circuitry **364** operates to: receive the peak cancellation pulse z(m) at the first terminal **366**; receive x(m)_DLY at the second terminal **368**; combine (e.g., add together) z(m) and x(m)_DLY; and provide a corrected transmit signal at the third terminal **370** responsive to the combination of z(m) and x(m)_DLY. In some examples, the peak cancellation circuitry **380** may include multiple peak cancellation resources that operate in parallel. In such examples, each of the peak cancellation resources may include a respective cancellation phasor control circuitry (e.g., the cancellation phasor control circuitry **114**A) and respective cancellation pulse generation circuitry (e.g., cancellation pulse generation circuitry **345**). Also, a scheduler may be used to manage resource-to-peak allocation to enable each of the peak cancellation resources to service a different detected peak. The cancellation pulses generated by the peak cancellation resources may be provided to the combine circuitry **364**. In such examples, the combine circuitry **364** may add the z(m) output from different peak cancellation resources (assigned to different peaks) to x(m)_DLY.
(49) In some examples, the additional CFR stages **372** operate to: receive the corrected transmit signal from the combine circuitry **364** at the first terminal **374**; iteratively perform one or more stages of CFR operations on the corrected transmit signal; and provide an iteratively corrected transmit signal at the second terminal **376**. For each stage of the additional CFR stages **372**, the operations described for the interpolation circuitry **302**, the interpolated envelope calculation circuitry **310**, the peak neighborhood analyzer **110**A, the peak detection circuitry **318**, the cancellation phasor control circuitry **114**A, the cancellation pulse generation circuitry **345**, the delay circuitry **356**, and the combine circuitry **364** are repeated. In some examples, peak detection/cancellation is performed using oversampled peak cancellation pulse to address hidden peaks. In different examples, oversampling ratios of 1, 2, 4, and/or 8 may be used for peak detection/cancellation. In some examples, the iteratively corrected transmit signal, resulting from operations of the additional CFR stages **372**, is provided to interpolation circuitry (e.g., the interpolation circuitry **220**) in preparation of other signal conditioning operations (e.g., DPD correction operations).
(50) FIG. **4** is a diagram showing other example CFR circuitry **400**. The CFR circuitry **400** is an example of at least some of the CFR circuitry **108** in FIG. **1**, at least some of the CFR circuitry **108**A in FIG. **2**, or at least some of the CFR circuitry **108**B in FIG. **3**. As shown, the CFR circuitry **400** includes a tapped delay line, a set of poly-phase filters **408**A to **408**M, interpolated envelope calculation circuitry **310**A, peak detection circuitry **318**A, and data/clock gating control circuitry **432**. The interpolated envelope calculation circuitry **310**A is an example of the interpolated envelope calculation circuitry **310** in FIG. **3**. The peak detection circuitry **318**A is an example of peak detection circuitry **318** in FIG. **3**.

(51) In the example of FIG. **3**, the tapped delay line **402** includes a first terminal **404** and a set of second terminals **406**A to **406**M. The each of the poly-phase filters of the set of poly-phase filters **408**A-**408**M includes a respective terminal of a set of first terminals **410**A to **410**M, a respective terminal of a set of second terminals **412**A to **412**M, and a respective terminal of a set of third terminals **414**A to **414**M. The interpolated envelope calculation circuitry **310**A has a set of first terminals **312**A to **312**M, the second terminal **314**, and a set of third terminals **316**A to **316**M. Each terminal of the set of first terminals **312**A to **312**M are examples of the first terminal **312** in FIG. **3**. Each terminal of the set of third terminals **316**A to **316**M is an example of the third terminal **316** in FIG. **3**. The peak detection circuitry **318**A has a set of first terminals **320**A to **320**M, the second terminal **322**, and the third terminal **324**. Each terminal of the set of first terminals **320**A to **320**M is an example of the first terminal **320** in FIG. **3**. The data/clock gating control circuitry **430** has a first terminal **434**, a set of second terminals **436**A to **436**M, a third terminal **438**, and a fourth terminal **440**.
(52) In the example of FIG. **4**, each terminal of the set of second terminals **406**A to **406**M of the tapped delay line **402** is coupled to a respective terminal of the set of first terminals **410**A to **410**N of the set of poly-phase filters **408**A to **408**M. Each terminal of the set of second terminals **412**A to **412**M of the poly-phase filters **408**A to **408**M is coupled to a respective terminal of the set of second terminals **436**A to **436**M of the data/clock gating control circuitry **432**. Each terminal of the set of third terminals **414**A to **414**M of the poly-phase filters **408**A to **408**M is coupled to a respective terminals of the set of first terminals **312**A to **312**M of the interpolated envelope calculation circuitry **310**A. The second terminal **314** of the interpolated envelope calculation circuitry **310**A is coupled to the third terminal **438** of the data/clock gating control circuitry **432**. Each terminal of the set of third terminal of the interpolated envelope calculation circuitry **310**A is coupled to a respect terminal of the set of first terminals **320**A to **320**M of the peak detection circuitry **318**A. The second terminal **322** of the peak detection circuitry **318**A is coupled to the fourth terminal **440** of the data/clock gating control circuitry **432**.
(53) The data/clock gating control circuitry **432** operates to: receive l(m) at the first terminal **434**; provide data/clock signals at the terminals of the set of second terminals **436**A to **436**M responsive to l(m) indicating a peak; provide data/clock signals at the third terminal **438** responsive to l(m) indicating a peak; and provide data/clock signals at the fourth terminal **440** responsive to l(m) indicating a peak.
(54) The tapped delay line **402** operates to: receive x(m) at the first terminal **404**; and provide delayed versions of x(m) to respective terminals of the set of second terminals **406**A to **406**M. In the example of FIG. **4**, the delayed versions of x(m) include x(m−1) to x(m−L), where L is the maximum sample lag related to the poly-phase filters **408**A to **408**M. The poly-phase filters **408**A to **408**M operate to: receive the delayed versions of x(m) at respective terminals of the set of first terminals **410**A to **410**M; selectively receive data/clock signals at respective terminals of the set of second terminals **412**A to **412**M responsive to the operations of the data/clock gating control circuitry **432** and l(m); and selectively provide interpolated samples (e.g., y(Mm) to y(Mm+M−1)) of x(m) at respective terminals of the set of third terminals **414**A to **414**M responsive to x(m) and the data/clock signals. In some examples, each terminal of the set of second terminals **406**A to **406**M of the tapped delay line **402** may include a separate terminal for each of the delayed versions of x(m) output by the tapped delay line **402**. Similarly, each terminal of the set of first terminals **410**A to **410**M of the poly-phase filters **408**A to **408**M may include a separate terminal for each of the delayed versions of x(m), which are received by each of the poly-phase filters **408**A to **408**M. In other words, as used herein, a “terminal” may support multiple inputs or outputs in series or in parallel.
(55) The interpolated envelope calculation circuitry **310**A operates to: receive the interpolated samples (e.g., y(Mm) to y(Mm+M−1)) of x(m) at the terminals of the set of first terminals **312**A to **312**M; selectively receive data/clocks at the second terminal **314** responsive to the operations of the

data/clock gating control circuitry **432** and l(m); calculate an envelope value for each of the interpolated samples of x(m); and selectively provide the calculated envelope values at respective terminals of the set of third terminals **316**A to **316**M responsive to the interpolated samples of x(m) and the data/clock signals.

(56) The peak detection circuitry **318**A operates to: receive the calculated envelope values at the terminals of the set of first terminals **320**A to **320**M; receive data/clock signals at the second terminal **322** responsive to the operations of the data/clock gating control circuitry **432** and l(m); selective compare the calculated envelope values to a peak limit to obtain comparison results and the data/clock signals; and selectively provide the comparison results to the third terminal **324** responsive to the comparison results.

(57) FIG. **5** is a graph **500** showing a complex envelope complementary cumulative distribution function (CCDF) of an orthogonal frequency division multiplexing (OFDM) signal. As shown in graph **500**, the probability of exceeding a peak limit decreases as the peak limit increases.

(58) FIG. **6**A is a graph **600** showing an example input signal envelope **602**, a peak absence prediction threshold, and a peak limit. In the example of FIG. **6**A, the value of the input signal envelope **602** varies as a function of time and includes five peaks P1 to P5. In the described examples, the peak absence prediction threshold is used to determine when at least some components of a CFR can be disabled or have their respective data/clocks gated. Based on the graph **600**, a peak neighborhood analyzer (e.g., the peak neighborhood analyzer **110** in FIGS. **1** and **2**, or the peak neighborhood analyzer **110**A in FIG. **3**) may determine that P1, P3, and P5 of the input signal envelope **602** are below the peak absence prediction threshold. Accordingly, at least some components of CFR circuitry (e.g., the poly-phase filters **408**A to **408**M, the interpolated envelope calculation circuitry **310** in FIG. **3**, the interpolated envelope calculation circuitry **310**A in FIG. **4**, the peak detection circuitry **318** in FIG. **3**, the peak detection circuitry **318**A in FIG. **4**, etc.) may be temporarily turned off or have their respective data/clocks temporarily gated. Also, based on the graph **600**, peak neighborhood analyzer may determine that P2 and P4 of the input signal envelope **602** are above the peak absence prediction threshold. Accordingly, the CFR circuitry is enabled during the intervals with P2 and P4. In the example of graph **600**, P2 is above the peak limit and P4 is below the peak limit. Accordingly, peak cancellation circuitry (e.g., the peak cancellation circuitry **380** in in FIG. **3**) may apply a correction to P2 and not P4. As described herein, the correction for P2 may involve: determining an excess peak portion relative to the peak limit; and applying a correction to P2 based on the excess peak portion. In some examples, the correction applied to P2 involves adding a cancellation pulse as described herein.

(59) FIG. **6**B is a diagram **610** showing an example complementary cumulative distribution function (CCDF) of a peak-to-average ratio (PAR) for a CFR input **612** and a CFR output **614**. In the diagram **610**, the probably of exceeding PAR for the CFR input **612** decreases as PAR increases. With CFR operations, the PAR of the CFR output **614** can be limited to a target PAR (e.g., 8 dB).

(60) In some examples, the described CFR techniques exploit the sparsity of peaks in 4G/5G transmit signal waveforms and related OFDM modulation, and temporal correlation in magnitude. In an example scenario, only ˜0.2% of the input samples exceed a peak limit of 8.5 dB, an example PAR target in 4G/5G wireless base stations. In different examples, the peak absence prediction threshold and the time window duration t may vary. In some examples, the peak absence prediction threshold and the time window duration t are selected to avoid missing peaks while some of the CFR circuitry is in the low power “inactive” state. With the described CFR techniques, significant power savings can be achieved without adding much overhead.

(61) FIG. **7** is a block diagram of example cancellation phasor control circuitry **114**B. The cancellation phasor control circuitry **114**B is an example of the cancellation phasor control circuitry **114** in FIG. **1**, or the cancellation phasor control circuitry **114**A in FIG. **3**. In the example of FIG. **7**, the cancellation phasor control circuitry **114**B has the first terminal **342** and the second terminal

344 described in FIG. 3. The cancellation phasor control circuitry 114B includes a delay line 702, the envelope calculation circuitry 346, excess peak calculation circuitry 708, and a multiplier 722. The delay line 702 has a first terminal 704 and a second terminal 706. The envelope calculation circuitry 346 has the first terminal 348 and the second terminal 350 described in FIG. 3. The excess peak calculation circuitry 708 has a first terminal 710, a second terminal 712, and a third terminal 714. The multiplier 722 has a first terminal 724, a second terminal 726, and a third terminal 728.

(62) In the example of FIG. 7, the first terminal 342 of the cancellation phasor control circuitry 114B is coupled to the first terminal of the delay line 702 and to the first terminal 348 of the envelope calculation circuitry 346. The second terminal 350 of the envelope calculation circuitry 346 is coupled to the first terminal of the excess peak calculation circuitry 708. The second terminal 712 of the excess peak calculation circuitry 708 is coupled to a peak limit source (not shown), which provides the peak limit or a related parameter. The peak limit source may be a register or other programmable storage unit. The third terminal 714 of the excess peak calculation circuitry 708 is coupled to a second terminal 726 of the multiplier 722. The first terminal 724 of the multiplier 722 is coupled to the second terminal 706 of the delay line 702. The third terminal 728 of the multiplier 722 is coupled to the second terminal 344 of the cancellation phasor control circuitry 114B.

(63) The delay line 702 operates to: receive a complex baseband interpolated sample y.sub.i corresponding to the detected peak at the first terminal 704; and provide a delayed version of y.sub.i at the second terminal 706. The envelope calculation circuitry 346 operates to: receive y.sub.i at the first terminal 348; calculate an envelope value of y.sub.i; and provide the calculated envelope value at the second terminal 350. The excess peak calculation circuitry 708 operates to: receive the calculated envelope value at the first terminal 710; receive the peak limit or related parameter at the second terminal 712; determine an excess peak amount responsive to the calculated envelope value and the peak limit; determine a scaling factor based on the excess peak amount; and provide the scaling factor at the third terminal 316. The multiplier 722 operates to receive the delayed version of the detected peak y.sub.i at the first terminal 724; receive the scaling factor at the second terminal 726; multiply the delayed version of y.sub.i by the scaling factor to generate a cancellation phasor for the detected peak; and provide the cancellation phasor at the third terminal 728. The second terminal 344 of the cancellation phasor control circuitry 114B receives the cancellation phasor.

(64) In some examples, the cancellation phasor control circuitry 114B operates to: perform an envelope calculation for a peak complex baseband interpolated sample y.sub.i (e.g., to determine |y.sub.i|.sup.2); calculate the excess peak to determine a scaling factor {1−(λ/|y.sub.i|)} from |y.sub.i|.sup.2 and λ; and perform scaling of y.sub.i responsive to the scaling factor to generate a cancellation phasor. In some examples, the cancellation phasor is given as α.sub.ie.sup.jΦ.sup.i, where e.sup.jΦ.sup.i is the phasor, and a; is a scaling applied to the phasor.

(65) With the cancellation phasor control circuitry 114B, explicit computation of the phase φ.sub.i from the complex sample y.sub.i and sin/cos computations are avoided. In some examples, the scaling factor is determined based on a small look-up table (LUT) to represent the function ƒ(x)=√{square root over (1/x)}, ∀x∈[1/16, 1]. As an option, the function may be coupled with linear interpolation between two adjacent values of the LUT indexed by the quantized value of |y.sub.i|.sup.2. After LUT look-up is performed, the output of the function ƒ(|y.sub.i|.sup.2) is multiplied by λ and is subtracted from the scaled output {λ*ƒ(|y.sub.i|.sup.2)} from unity, which reduces complexity compared to computation of the phase φ.sub.i from the complex sample y.sub.i and sin/cos computations. With the described examples, power consumption of CFR peak detection is reduced. Also, the modified cancellation phasor computation scheme results in digital area savings. These benefits are achieved without affecting CFR performance.

(66) In some examples, a circuit (e.g., the circuit 100 in FIG. 1) includes CFR circuitry (e.g., the CFR circuitry 108 in FIG. 1, the CFR circuitry 108A in FIG. 2, the CFR circuitry 108B in FIG. 3,

of the CFR circuitry **400** in FIG. **4**). The CFR circuitry includes: a peak neighborhood analyzer (e.g., the peak neighborhood analyzer **110**A in FIG. **3**) having a first terminal (e.g., the first terminal **326** in FIG. **3**) and a second terminal (e.g., the second terminal **328** in FIG. **3**); peak detection circuitry (e.g., the peak detection circuitry **318** in FIG. **3**) having a first terminal (e.g., the first terminal **320** in FIG. **3**), a second terminal (e.g., the second terminal **322** in FIG. **3**), and a third terminal (e.g., the third terminal **324** in FIG. **3**); and a controller (e.g., the data/clock gating control circuitry **432** in FIG. **4**) having a first terminal (e.g., the first terminal **434** in FIG. **4**) and a second terminal (e.g., the fourth terminal **440** in FIG. **4**). The first terminal of the controller is coupled to the second terminal of the peak neighborhood analyzer. The second terminal of the controller is coupled to the second terminal of the peak detection circuitry. The peak neighborhood analyzer is configured to: receive an input signal (e.g., x(m) herein) at the first terminal of the peak neighborhood analyzer; analyze the input signal to determine whether a peak larger than a target threshold (e.g., the peak absence prediction threshold herein) is expected within an interval; and provide a first control signal (e.g., l(m) herein) at the second terminal of the peak neighborhood analyzer responsive to determining that a peak larger than the target threshold is expected within the interval. The controller is configured to: receive the first control signal at the first terminal of the controller; and gate a clock or data to the peak detection circuitry responsive to the first control signal.

(67) In some examples, the peak neighborhood analyzer includes envelope calculation circuitry (e.g., the envelope calculation circuitry **330** in FIG. **3**) and peak absence prediction circuitry (e.g., the peak absence prediction circuitry **112**A in FIG. **3**). The envelope calculation circuitry has a first terminal (e.g., the first terminal **332** in FIG. **3**) and a second terminal (e.g., the second terminal **334** in FIG. **3**). The peak absence prediction circuitry has a first terminal (e.g., the first terminal **338** in FIG. **3**) and a second terminal (e.g., the second terminal **340** in FIG. **3**). The first terminal of the envelope calculation circuitry is coupled to the first terminal of the peak neighborhood analyzer. The second terminal of the envelope calculation circuitry is coupled to the first terminal of the peak absence prediction circuitry. The second terminal of the peak absence prediction circuitry coupled to the second terminal of the peak neighborhood analyzer.

(68) In some examples, the peak neighborhood analyzer is configured to: receive input signal samples at the first terminal of the peak neighborhood analyzer; analyze the input signal samples to determine whether a peak larger than a target threshold is expected within an interval; and provide a control signal at the second terminal of the peak neighborhood analyzer responsive to determining that a peak larger than the target threshold is not expected within the interval. In some examples, the envelope calculation circuitry is configured to calculate envelope values of the input signal samples, and the peak absence prediction circuitry is configured to: receive the envelope values at the first terminal of the peak absence prediction circuitry; compare the envelope values to the target threshold to obtain comparison results; and provide the control signal at the first terminal of the peak absence prediction circuitry responsive to the comparison results indicating the envelope values stay below the target threshold for at least a threshold number of input signal samples.

(69) In some examples, the envelope calculation circuitry (e.g., the envelope calculation circuitry **330** in FIG. **3**) is configured to: receive input signal samples (e.g., x(m) herein) at the first terminal of the envelope calculation circuitry, the input signal samples associated with a baseband interface rate; calculate envelope values (e.g., |x(m)|.sup.2 herein) of the input signal samples; and provide the envelope values at the second terminal of the envelope calculation circuitry. In some examples, the peak absence prediction circuitry is configured to: receive the envelope values at the first terminal of the peak absence prediction circuitry; compare the envelope values to the target threshold to obtain comparison results; and provide the first control signal responsive to the comparison results indicating the envelope values stay below the target threshold for at least a threshold number of input signal samples.

(70) In some examples, the controller has a third terminal (e.g., the third terminal **438** in FIG. **4**), the CFR circuitry includes interpolated envelope calculation circuitry (e.g., the interpolated envelope calculation circuitry **310** in FIG. **3**) having a first terminal (e.g., the first terminal **312** in FIG. **3**), a second terminal (e.g., the second terminal **314** in FIG. **3**) and a third terminal (e.g., the third terminal **316** in FIG. **3**). The second terminal of the interpolated envelope calculation circuitry is coupled to the third terminal of the controller. The third terminal of the interpolated envelope calculation circuitry is coupled to the first terminal of the peak detection circuitry. In such examples, the controller is configured to: receive the first control signal at the first terminal of the controller; and gate a clock or data to the interpolated envelope calculation circuitry responsive to the first control signal.
(71) In some examples, the controller has a fourth terminal (e.g., the set of second terminals **436**A to **436**M in FIG. **4**), and the CFR circuitry includes interpolation circuitry (e.g., the interpolation circuitry **302** in FIG. **3**) having a first terminal (e.g., the first terminal **304** in FIG. **3**), a second terminal (e.g., the second terminal **306** in FIG. **3**) and a third terminal (e.g., the third terminal **308** in FIG. **3**). The second terminal of the interpolation circuitry is coupled to the fourth terminal of the controller. The third terminal of the interpolation circuitry is coupled to the first terminal of the interpolated envelope calculation circuitry. In such examples, the controller is configured to: receive the first control signal at the first terminal of the controller; and gate a clock or data to the interpolation circuitry responsive to the first control signal.
(72) In some examples, the fourth terminal of the controller is one terminal of a set of fourth terminals (e.g., the set of second terminals **436**A to **436**M in FIG. **4**). The interpolation circuitry includes a set of poly-phase filters (e.g., the set of poly-phase filters **408**A to **408**M in FIG. **4**). Each poly-phase filter of the set of poly-phase filters has a first terminal (e.g., a respective terminal of the set of first terminals **410**A to **410**M in FIG. **4**), a second terminal (e.g., a respective terminal of the set of second terminals **412**A to **412**M in FIG. **4**), and a third terminal (e.g., a respective terminal of the set of third terminals **414**A to **414**M in FIG. **4**). Each second terminal of each respective poly-phase filter is coupled to a respective terminal of the set of fourth terminals. In such examples, the controller is configured to: receive the first control signal at the first terminal of the controller; and gate a clock or data to each poly-phase filter of the set of poly-phase filters responsive to the first control signal.
(73) In some examples, the CFR circuitry includes peak cancellation circuitry (e.g., the peak cancellation circuitry **380** in FIG. **3**) having a first terminal (e.g., the first terminal **382** in FIG. **3**), a second terminal (e.g., the second terminal **384** in FIG. **3**), and a third terminal (e.g., the third terminal **386** in FIG. **3**). In such examples, the first terminal of the peak cancellation circuitry is coupled to the third terminal (e.g., the third terminal **324** in FIG. **3**) of the peak detection circuitry. The peak detection circuitry is configured to detect when a peak of the input signal is greater than a peak limit, the peak limit greater than the target threshold. The peak cancellation circuitry is configured to: calculate an excess peak portion of the detected peak relative to the peak limit; and apply a correction to the detected peak based on the excess peak portion. In some examples, applying a correction to the detected peak is based on adding a cancellation pulse to the detected peak.
(74) In some examples, the CFR circuitry includes peak cancellation circuitry (e.g., the peak cancellation circuitry **380** in FIG. **3**) having a first terminal (e.g., the first terminal **382** in FIG. **3**), a second terminal (e.g., the second terminal **384** in FIG. **3**), and a third terminal (e.g., the third terminal **386** in FIG. **3**). The first terminal of the peak cancellation circuitry is coupled to the third terminal (e.g., the third terminal **324** in FIG. **3**) of the peak detection circuit. The second terminal of the peak cancellation circuitry is coupled to the first terminal (e.g., the first terminal **326** in FIG. **3**) of the peak neighborhood analyzer. In such examples, the peak cancellation circuitry may include: envelope calculation circuitry (e.g., the envelope calculation circuitry **346** in FIGS. **3** and **7**); excess peak calculation circuitry (e.g., the excess peak calculation circuitry **708** in FIG. **7**); a delay line

(e.g., the delay line **702** in FIG. **7**); and a multiplier (e.g., the multiplier **722** in FIG. **7**). The envelope calculation circuitry has a first terminal (e.g., the first terminal **348** in FIGS. **3** and **7**) and a second terminal (e.g., the second terminal **350** in FIGS. **3** and **7**). The first terminal of the envelope calculation circuitry is coupled to the first terminal of the peak cancellation circuitry. The excess peak calculation circuitry has a first terminal (e.g., the first terminal **710** in FIG. **7**) and a second terminal (e.g., the third terminal **714** in FIG. **7**). The first terminal of the excess peak calculation circuitry is coupled to the second terminal of the envelope calculation circuitry. The delay line has a first terminal (e.g., the first terminal **704** in FIG. **7**) and a second terminal (e.g., the second terminal **706** in FIG. **7**). The first terminal of the delay line is coupled to the first terminal of the peak cancellation circuitry. The multiplier has a first terminal (e.g., the first terminal **724** in FIG. **7**), a second terminal (e.g., the second terminal **726** in FIG. **7**), and a third terminal (e.g., the third terminal **728** in FIG. **7**). The first terminal of the multiplier is coupled to the second terminal of the delay line. The second terminal of the multiplier is coupled to the second terminal of the excess peak calculation circuitry.

(75) In some examples, the peak cancellation circuitry includes: delay circuitry (e.g., the delay circuitry **356** in FIG. **3**); and combine circuitry (e.g., the combine circuitry **364** in FIG. **3**). The delay circuitry has a first terminal (e.g., the first terminal **358** in FIG. **3**) and a second terminal (e.g., the third terminal **362** in FIG. **3**). The first terminal of the delay circuitry is coupled to the second terminal of the peak cancellation circuitry. The combine circuitry has a first terminal (e.g., the first terminal **366** in FIG. **3**), a second terminal (e.g., the second terminal **368** in FIG. **3**), and a third terminal (e.g., the third terminal **370** in FIG. **3**). The first terminal of the combine circuitry is coupled to the third terminal of the multiplier. The second terminal of the combine circuitry is coupled to the second terminal of the delay circuitry. The third terminal of the combine circuitry is coupled to the third terminal of the peak cancellation circuitry.

(76) In some examples, a transmitter (e.g., the transmitter **102** in FIG. **1**, or the transmitter circuitry **208** in FIG. **2**) includes CFR circuitry (e.g., the CFR circuitry **108** in FIG. **1**, the CFR circuitry **108**A in FIG. **2**, the CFR circuitry **108**B in FIG. **3**, of the CFR circuitry **400** in FIG. **4**). The CFR circuitry includes: peak detection circuitry (e.g., the peak detection circuitry **318** in FIG. **3**); and peak cancellation circuitry (e.g., the peak cancellation circuitry **380** in FIG. **3**. The peak detection circuitry has a first terminal (e.g., the first terminal **320** in FIG. **3**), a second terminal (e.g., the second terminal **322** in FIG. **3**), and a third terminal (e.g., the third terminal **324** in FIG. **3**). The peak cancellation circuitry has a first terminal (e.g., the first terminal **382** in FIG. **3**), a second terminal (e.g., the second terminal **384** in FIG. **3**), and a third terminal (e.g., the third terminal **386** in FIG. **3**). The first terminal of the peak cancellation circuitry is coupled to the third terminal of the peak detection circuitry.

(77) In some transmitter examples, the peak cancellation circuitry may include: envelope calculation circuitry (e.g., the envelope calculation circuitry **346** in FIGS. **3** and **7**); excess peak calculation circuitry (e.g., the excess peak calculation circuitry **708** in FIG. **7**); a delay line (e.g., the delay line **702** in FIG. **7**); and a multiplier (e.g., the multiplier **722** in FIG. **7**). The envelope calculation circuitry has a first terminal (e.g., the first terminal **348** in FIGS. **3** and **7**) and a second terminal (e.g., the second terminal **350** in FIGS. **3** and **7**). The first terminal of the envelope calculation circuitry is coupled to the first terminal of the peak cancellation circuitry. The excess peak calculation circuitry has a first terminal (e.g., the first terminal **710** in FIG. **7**) and a second terminal (e.g., the third terminal **714** in FIG. **7**). The first terminal of the excess peak calculation circuitry is coupled to the second terminal of the envelope calculation circuitry. The delay line has a first terminal (e.g., the first terminal **704** in FIG. **7**) and a second terminal (e.g., the second terminal **706** in FIG. **7**). The first terminal of the delay line is coupled to the first terminal of the peak cancellation circuitry. The multiplier has a first terminal (e.g., the first terminal **724** in FIG. **7**), a second terminal (e.g., the second terminal **726** in FIG. **7**), and a third terminal (e.g., the third terminal **728** in FIG. **7**). The first terminal of the multiplier is coupled to the second terminal of the

delay line. The second terminal of the multiplier is coupled to the second terminal of the excess peak calculation circuitry.
(78) In some transmitter examples, the peak cancellation circuitry includes: cancellation pulse generation circuitry (e.g., the cancellation pulse generation circuitry **345** in FIG. **3**); delay circuitry (e.g., the delay circuitry **356** in FIG. **3**); and combine circuitry (e.g., the combine circuitry **364** in FIG. **3**). The cancellation pulse generation circuitry has a first terminal (e.g., the first terminal **347** in FIG. **3**) and a second terminal (e.g., the second terminal **349** in FIG. **3**). The first terminal of the cancellation pulse generation circuitry is coupled to the third terminal of the multiplier. The delay circuitry has a first terminal (e.g., the first terminal **358** in FIG. **3**) and a second terminal (e.g., the third terminal **362** in FIG. **3**). The first terminal of the delay circuitry is coupled to the second terminal of the peak cancellation circuitry. The combine circuitry has a first terminal (e.g., the first terminal **366** in FIG. **3**), a second terminal (e.g., the second terminal **368** in FIG. **3**), and a third terminal (e.g., the third terminal **370** in FIG. **3**). The first terminal of the combine circuitry is coupled to the second terminal of the cancellation pulse generation circuitry. The second terminal of the combine circuitry is coupled to the second terminal of the delay circuitry. The third terminal of the combine circuitry is coupled to the third terminal of the peak cancellation circuitry.
(79) In some transmitter examples, the CFR circuitry includes: a peak neighborhood analyzer (e.g., the peak neighborhood analyzer **110** in FIG. **1**, or the peak neighborhood analyzer **110**A in FIG. **3**); and a controller (e.g., the data/clock gating control circuitry **432** in FIG. **4**). The peak neighborhood analyzer has a first terminal (e.g., the first terminal **326** in FIG. **3**) and a second terminal (e.g., the second terminal **328** in FIG. **3**). The controller has a first terminal (e.g., the first terminal **434** in FIG. **4**) and a second terminal (e.g., the fourth terminal **440** in FIG. **4**). The first terminal of the controller is coupled to the second terminal of the peak neighborhood analyzer. The second terminal of the controller is coupled to the second terminal of the peak detection circuitry. In some transmitter examples, the peak neighborhood analyzer is configured to: receive input signal samples at the first terminal of the peak neighborhood analyzer; analyze the input signal samples to determine whether a peak larger than a target threshold is expected within an interval; and provide a control signal at the second terminal of the peak neighborhood analyzer responsive to determining that a peak larger than the target threshold is not expected within the interval.
(80) In some transmitter examples, the peak neighborhood analyzer includes envelope calculation circuitry (e.g., the envelope calculation circuitry **330** in FIG. **3**) and peak absence prediction circuitry (e.g., the peak absence prediction circuitry **112** in FIG. **1**, or the peak absence prediction circuitry **112**A in FIG. **3**). The envelope calculation circuitry has a first terminal (e.g., the first terminal **332** in FIG. **3**) and second terminal (e.g., the second terminal **334** in FIG. **3**). The peak absence prediction circuitry has a first terminal (e.g., the first terminal **338** in FIG. **3**) and a second terminal (e.g., the second terminal **340** in FIG. **3**). The first terminal of the envelope calculation circuitry is coupled to the first terminal of the peak neighborhood analyzer. The second terminal of the envelope calculation circuitry is coupled to the first terminal of the peak absence prediction circuitry. The second terminal of the peak absence prediction circuitry is coupled to the second terminal of the peak neighborhood analyzer. The envelope calculation circuitry is configured to calculate envelope values of the input signal samples. The peak absence prediction circuitry configured to: receive the envelope values at the first terminal of the peak absence prediction circuitry; compare the envelope values to the target threshold to obtain comparison results; and provide the control signal at the first terminal of the peak absence prediction circuitry responsive to the comparison results indicating the envelope values stay below the target threshold for at least a threshold number of input signal samples.
(81) In some examples, the controller has a third terminal (e.g., the third terminal **438** in FIG. **4**), and the CFR circuitry includes interpolated envelope calculation circuitry (e.g., the interpolated envelope calculation circuitry **310** in FIG. **3**, or the interpolated envelope calculation circuitry **310**A in FIG. **4**). The interpolated envelope calculation circuitry has a first terminal (e.g., the first

terminal **312** in FIG. **3**), a second terminal (e.g., the second terminal **314** in FIG. **3**), and a third terminal (e.g., the third terminal **316** in FIG. **3**). The second terminal of the interpolated envelope calculation circuitry is coupled to the third terminal of the controller. The third terminal of the interpolated envelope calculation circuitry is coupled to the first terminal of the peak detection circuitry. In such examples, the controller is configured to: receive the control signal at the first terminal of the controller; and gate a clock or data to the interpolated envelope calculation circuitry responsive to the control signal.

(82) In some examples, the controller has a fourth terminal (e.g., the second of second terminals **436**A to **436**M). The CFR circuitry includes interpolation circuitry (e.g., the interpolation circuitry **302** in FIG. **3**, or the poly-phase filters **408**A to **408**M in FIG. **4**). The interpolation circuitry has a first terminal (e.g., the first terminal **304** in FIG. **3**, or related set of first terminals **410**A to **410**M in FIG. **4**), a second terminal (e.g., the second terminal **306** in FIG. **3**, or the related set of second terminals **412**A to **412**M in FIG. **4**), and a third terminal (e.g., the third terminal **308** in FIG. **3**, or the set of third terminals **414**A to **414**M in FIG. **4**). In such examples, the second terminal of the interpolation circuitry is coupled to the fourth terminal of the controller. The third terminal of the interpolation circuitry is coupled to the first terminal of the interpolated envelope calculation circuitry. The controller is configured to: receive the control signal at the first terminal of the controller; and gate a clock or data to the interpolation circuitry responsive to the control signal.

(83) In some examples, the transmitter may be a stand-alone integrated circuit (IC) product or may be part of a larger IC. In some examples, an IC may include the transmitter and a baseband processor that provides transmit data to the transmitter. In some examples, an IC may include the transmitter and analog front-end (AFE) circuitry.

(84) FIGS. **8**, **9**, and **10** are flowcharts of example CFR methods **800**, **900**, and **1000**. The CFR methods **800**, **900**, and **1000** may be performed by CFR circuitry (e.g., the CFR circuitry **108** in FIG. **1**, the CFR circuitry **108**A in **2**, the CFR circuitry **108**B in FIG. **3**, the CFR circuitry **400** in FIG. **4**, and/or the cancellation phasor control circuitry **114**B in FIG. **7**. In FIG. **8**, the CFR method **800** includes receiving a transmit signal at block **802**. Without limitation, the transmit signal may be a 4G or 5G transmit signal with OFDM modulation. At block **804**, a peak is detected in the transmit signal. At block **806**, an express peak portion of the detected peak is calculated relative to a peak limit. At block **808**, a correction is applied to the detected peak based on the excess peak portion. At block **810**, an updated transmit signal with the corrected peak is provided.

(85) In FIG. **9**, the CFR method **900** includes receiving a transmit signal at block **902**. Without limitation, the transmit signal may be a 4G or 5G transmit signal with OFDM modulation. At block **904**, peak detection is selectively enabled based on a peak absence prediction for the transmit signal. When enabled (e.g., in response to peak absence prediction results indicating a peak is expected), peak detection is performed by comparing a peak of the transmit signal to a peak limit at block **906**. At block **908**, a correction is applied if the detected peak is greater than the peak limit. At block **910**, an updated transmit signal based on the correction is provided.

(86) In FIG. **10**, the CFR method **1000** includes receiving a transmit signal at block **1002**. Without limitation, the transmit signal may be a 4G or 5G transmit signal with OFDM modulation. At block **1004**, peak detection is selectively enabled based on a peak absence prediction for the transmit signal. When enabled (e.g., in response to peak absence prediction results indicating a peak is expected), peak detection is performed by comparing a peak of the transmit signal to a peak limit at block **1006**. At block **1008**, an excess peak portion of the peak relative to the peak limit is calculated. At block **1010**, a correction is applied to the peak based on the excess peak portion. At block **1012**, an updated transmit signal based on the correction is provided.

(87) In some examples, the calculating the excess peak portion in the CFR methods **800** and **1000** may include: calculating an envelope value for an interpolated transmit signal sample; comparing the envelope value to the peak limit to obtain comparison results; and determine the excess peak portion based on the comparison results. In some examples, performing peak absence detection in

CFR methods **900** and **1000** may include: receiving transmit signal samples for an interval of the transmit signal; calculating envelope values for the transmit signal samples; comparing the envelope values to a threshold to obtain comparison results, the threshold less than the peak limit; and in response to the comparison results indicating there is no peak larger than the threshold within the interval, providing a control signal. In such examples, the CFR method **900** and/or the CFR method **1000** may include: disabling peak detection circuitry used to perform the peak detection responsive to the control signal; disabling interpolated envelope calculation circuitry used to provide envelope values to the peak detection circuitry responsive to the control signal; and/or disabling interpolation circuitry used to provide interpolated transmit signal samples to the interpolated envelope calculation circuitry responsive to the control signal.

(88) In some examples, CFR circuitry includes: envelope computation to determine the magnitude of a complex input signal at a baseband interface rate; peak absence prediction that indicates the absence of peaks in the vicinity of the current input sample; and data/clock gating to transition interpolation components and/or CFR components between an active state and an inactive (low power) state responsive to peak absence prediction results. In some examples, peak absence prediction is based on comparing the input signal magnitude against a threshold, which is lower than the peak limit value by an adjustable offset. In some examples, peak absence prediction is based on absence of threshold crossing for an adjustable time interval or window. In some examples, interpolation components (e.g., poly-phase filters), interpolated envelope computation circuitry and peak detection circuitry are transitioned between an active state and an inactive state (e.g., by appropriate data/CLK gating) based on the peak absence prediction results. In some examples, cancellation phasor control includes: calculation of an excess peak portion of a detected peak based on the envelope value of the peak complex baseband interpolated sample and a peak limit; and scaling the peak complex baseband interpolated sample responsive to the excess peak portion to generate a cancellation phasor. In some examples, a peak cancellation waveform is scaled responsive to the cancellation phasor to generate a peak cancellation pulse. The peak cancellation pulse is added to a delayed (latency-matched) version of the transmit signal, resulting in a corrected transmit signal. In some examples, the excess peak portion is calculated based on an inverse square-root computation.

(89) In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

(90) Also, in this description, the recitation “based on” means “based at least in part on.” Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

(91) A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

(92) As used herein, the terms “terminal”, “node”, “interconnection”, “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

(93) A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example,

a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.
(94) Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.
(95) While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other examples, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated circuit. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.
(96) Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within +/−10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.
(97) Modifications are possible in the described examples, and other examples are possible, within the scope of the claims.

Claims

1. A circuit comprising: crest factor reduction (CFR) circuitry including: a peak neighborhood analyzer having a first terminal and a second terminal; peak detection circuitry having a first terminal, a second terminal, and a third terminal; and a controller having a first terminal and a second terminal, the first terminal of the controller coupled to the second terminal of the peak neighborhood analyzer, the second terminal of the controller coupled to the second terminal of the peak detection circuitry, wherein the peak neighborhood analyzer is configured to: receive an input signal at the first terminal of the peak neighborhood analyzer; analyze the input signal to determine whether a peak larger than a target threshold is expected within an interval; and provide a first control signal at the second terminal of the peak neighborhood analyzer responsive to determining that a peak larger than the target threshold is expected within the interval, and wherein the controller is configured to: receive the first control signal at the first terminal of the controller; and gate a clock or data to the peak detection circuitry responsive to the first control signal.
2. The circuit of claim 1, wherein the peak neighborhood analyzer includes envelope calculation circuitry and peak absence prediction circuitry, the envelope calculation circuitry having a first

terminal and a second terminal, the peak absence prediction circuitry having a first terminal and a second terminal, the first terminal of the envelope calculation circuitry coupled to the first terminal of the peak neighborhood analyzer, the second terminal of the envelope calculation circuitry coupled to the first terminal of the peak absence prediction circuitry, the second terminal of the peak absence prediction circuitry coupled to the second terminal of the peak neighborhood analyzer.

3. The circuit of claim 2, wherein the envelope calculation circuitry is configured to: receive input signal samples at the first terminal of the envelope calculation circuitry, the input signal samples associated with a baseband interface rate; calculate envelope values of the input signal samples; and provide the envelope values at the second terminal of the envelope calculation circuitry.

4. The circuit of claim 3, wherein the peak absence prediction circuitry is configured to: receive the envelope values at the first terminal of the peak absence prediction circuitry; compare the envelope values to the target threshold to obtain comparison results; and provide the first control signal responsive to the comparison results indicating the envelope values stay below the target threshold for at least a threshold number of input signal samples.

5. The circuit of claim 1, wherein the controller has a third terminal, the CFR circuitry includes interpolated envelope calculation circuitry having a first terminal, a second terminal, and a third terminal, the second terminal of the interpolated envelope calculation circuitry coupled to the third terminal of the controller, the third terminal of the interpolated envelope calculation circuitry coupled to the first terminal of the peak detection circuitry, and the controller is configured to: receive the first control signal at the first terminal of the controller; and gate a clock or data to the interpolated envelope calculation circuitry responsive to the first control signal.

6. The circuit of claim 5, wherein the controller has a fourth terminal, the CFR circuitry includes interpolation circuitry having a first terminal, a second terminal, and a third terminal, the second terminal of the interpolation circuitry coupled to the fourth terminal of the controller, the third terminal of the interpolation circuitry coupled to the first terminal of the interpolated envelope calculation circuitry, and the controller is configured to: receive the first control signal at the first terminal of the controller; and gate a clock or data to the interpolation circuitry responsive to the first control signal.

7. The circuit of claim 6, wherein the fourth terminal of the controller is one terminal of a set of fourth terminals of the controller, the interpolation circuitry includes a set of poly-phase filters, each poly-phase filter of the set of poly-phase filters having a first terminal, a second terminal, and a third terminal, each second terminal of each respective poly-phase filter coupled to a respective terminal of the set of fourth terminals, and the controller is configured to: receive the first control signal at the first terminal of the controller; and gate a clock or data to each poly-phase filter of the set of poly-phase filters responsive to the first control signal.

8. The circuit of claim 1, wherein the CFR circuitry includes peak cancellation circuitry having a first terminal, a second terminal, and a third terminal, the first terminal of the peak cancellation circuitry is coupled to the third terminal of the peak detection circuitry, the peak detection circuitry is configured to detect when a peak of the input signal is greater than a peak limit, the peak limit greater than the target threshold, and the peak cancellation circuitry is configured to: calculate an excess peak portion of the detected peak relative to the peak limit; and apply a correction to the detected peak based on the excess peak portion.

9. The circuit of claim 1, wherein the CFR circuitry includes peak cancellation circuitry having a first terminal, a second terminal, and a third terminal, the first terminal of the peak cancellation circuitry coupled to the third terminal of the peak detection circuitry, the second terminal of the peak cancellation circuitry coupled to the first terminal of the peak neighborhood analyzer, and the peak cancellation circuitry including: envelope calculation circuitry having a first terminal and a second terminal, the first terminal of the envelope calculation circuitry coupled to the first terminal of the peak cancellation circuitry; excess peak calculation circuitry having a first terminal and a

second terminal, the first terminal of the excess peak calculation circuitry coupled to the second terminal of the envelope calculation circuitry; a delay line having a first terminal and a second terminal, the first terminal of the delay line coupled to the first terminal of the peak cancellation circuitry; and a multiplier having a first terminal, a second terminal, and a third terminal, the first terminal of the multiplier coupled to the second terminal of the delay line, the second terminal of the multiplier coupled to the second terminal of the excess peak calculation circuitry.

10. The circuit of claim 9, wherein the peak cancellation circuitry includes: cancellation pulse generation circuitry having a first terminal and a second terminal, the first terminal of the cancellation pulse generation circuitry coupled to the third terminal of the multiplier; delay circuitry having a first terminal and a second terminal, the first terminal of the delay circuitry coupled to the second terminal of the peak cancellation circuitry; and combine circuitry having a first terminal, a second terminal, and a third terminal, the first terminal of the combine circuitry coupled to the second terminal of the cancellation pulse generation circuitry, the second terminal of the combine circuitry coupled to the second terminal of the delay circuitry, and the third terminal of the combine circuitry coupled to the third terminal of the peak cancellation circuitry.

11. A transmitter comprising: crest factor reduction (CFR) circuitry including: peak detection circuitry having a first terminal, a second terminal, and a third terminal; and peak cancellation circuitry having a first terminal, a second terminal, and a third terminal, the first terminal of the peak cancellation circuitry coupled to the third terminal of the peak detection circuitry, the peak cancellation circuitry including: envelope calculation circuitry having a first terminal and a second terminal, the first terminal of the envelope calculation circuitry coupled to the first terminal of the peak cancellation circuitry; excess peak calculation circuitry having a first terminal and a second terminal, the first terminal of the excess peak calculation circuitry coupled to the second terminal of the envelope calculation circuitry; a delay line having a first terminal and a second terminal, the first terminal of the delay line coupled to the first terminal of the peak cancellation circuitry; and a multiplier having a first terminal, a second terminal, and a third terminal, the first terminal of the multiplier coupled to the second terminal of the delay line, the second terminal of the multiplier coupled to the second terminal of the excess peak calculation circuitry.

12. The transmitter of claim 11, wherein the peak cancellation circuitry includes: cancellation pulse generation circuitry having a first terminal and a second terminal, the first terminal of the cancellation pulse generation circuitry coupled to the third terminal of the multiplier; delay circuitry having a first terminal and a second terminal, the first terminal of the delay circuitry coupled to the second terminal of the peak cancellation circuitry; and combine circuitry having a first terminal, a second terminal, and a third terminal, the first terminal of the combine circuitry coupled to the second terminal of the cancellation pulse generation circuitry, the second terminal of the combine circuitry coupled to the second terminal of the delay circuitry, and the third terminal of the combine circuitry coupled to the third terminal of the peak cancellation circuitry.

13. The transmitter of claim 11, wherein the CFR circuitry includes: a peak neighborhood analyzer having a first terminal and a second terminal; and a controller having a first terminal and a second terminal, the first terminal of the controller coupled to the second terminal of the peak neighborhood analyzer, the second terminal of the controller coupled to the second terminal of the peak detection circuitry, the peak neighborhood analyzer configured to: receive input signal samples at the first terminal of the peak neighborhood analyzer; analyze the input signal samples to determine whether a peak larger than a target threshold is expected within an interval; and provide a control signal at the second terminal of the peak neighborhood analyzer responsive to determining that a peak larger than the target threshold is not expected within the interval.

14. The transmitter of claim 13, wherein the peak neighborhood analyzer includes envelope calculation circuitry and peak absence prediction circuitry, the envelope calculation circuitry having a first terminal and second terminal, the peak absence prediction circuitry having a first terminal and a second terminal, the first terminal of the envelope calculation circuitry coupled to

the first terminal of the peak neighborhood analyzer, the second terminal of the envelope calculation circuitry coupled to the first terminal of the peak absence prediction circuitry, the second terminal of the peak absence prediction circuitry coupled to the second terminal of the peak neighborhood analyzer, the envelope calculation circuitry configured to calculate envelope values of the input signal samples, and the peak absence prediction circuitry configured to: receive the envelope values at the first terminal of the peak absence prediction circuitry; compare the envelope values to the target threshold to obtain comparison results; and provide the control signal at the first terminal of the peak absence prediction circuitry responsive to the comparison results indicating the envelope values stay below the target threshold for at least a threshold number of input signal samples.

15. The transmitter of claim 13, wherein the controller has a third terminal, the CFR circuit further comprises interpolated envelope calculation circuitry having a first terminal, a second terminal, and a third terminal, the second terminal of the interpolated envelope calculation circuitry coupled to the third terminal of the controller, the third terminal of the interpolated envelope calculation circuitry coupled to the first terminal of the peak detection circuitry, and the controller is configured to: receive the control signal at the first terminal of the controller; and gate a clock or data to the interpolated envelope calculation circuitry responsive to the control signal.

16. The transmitter of claim 15, wherein the controller has a fourth terminal, the CFR circuitry includes interpolation circuitry having a first terminal, a second terminal, and a third terminal, the second terminal of the interpolation circuitry coupled to the fourth terminal of the controller, the third terminal of the interpolation circuitry coupled to the first terminal of the interpolated envelope calculation circuitry, and the controller is configured to: receive the control signal at the first terminal of the controller; and gate a clock or data to the interpolation circuitry responsive to the control signal.

17. A method comprising: receiving a transmit signal; selectively enabling peak detection based on a peak absence prediction for the transmit signal; when enabled, performing peak detection by comparing a peak of the transmit signal to a peak limit; calculating an excess peak portion of the peak relative to the peak limit; applying a correction to the peak based on the excess peak portion; and providing an updated transmit signal with the corrected peak.

18. The method of claim 17, wherein calculating the excess peak portion includes: calculating an envelope value for an interpolated transmit signal sample; comparing the envelope value to the peak limit to obtain comparison results; and determine the excess peak portion based on the comparison results.

19. The method of claim 17, wherein performing peak absence detection includes: receiving transmit signal samples for an interval of the transmit signal; calculating envelope values for the transmit signal samples; comparing the envelope values to a threshold to obtain comparison results, the threshold less than the peak limit; and in response to the comparison results indicating there is no peak larger than the threshold within the interval, providing a control signal, and the method further comprises disabling peak detection circuitry used to perform the peak detection responsive to the control signal.

20. The method of claim 19, further comprising disabling interpolated envelope calculation circuitry used to provide envelope values to the peak detection circuitry responsive to the control signal.